

Claire Tsui

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EDUCATION

University of California, San Diego

Sept 2021 - Jun 2025

Bachelor of Science in Mathematics (Applied), Minor in Computer Science

TECHNICAL SKILLS

Programming Languages: C#, Java, C++, Python, R, SQL, PHP, HTML/CSS/JS, Matlab, Bash

Frameworks & Tools: ASP.NET Core, React.js, Tailwind CSS, MSSQL, DBeaver, Postman, Docker, LaTeX

Languages: English, Mandarin (Fluent); Spanish (Basic)

WORK EXPERIENCE

GIGABYTE

Taipei, Taiwan

Software Engineer (Full-stack Developer)

Jul 2025 - present

- Led the redesign and redevelopment of internal web systems using ASP.NET Core (MVC), refactoring legacy code and modularizing service components to improve scalability, readability, and long-term maintainability
- Designed, implemented, and optimized 25+ APIs for real-time data retrieval; validated endpoints with Postman and improved query logic, reducing average API response time by ~30%
- Collaborated closely with frontend developers, product managers, and users to align feature requirements, conduct code reviews, and coordinate cross-team deployments, ensuring smooth integration and on-time delivery
- Managed and queried large relational databases using MSSQL and DBeaver, contributing to backend data operations and supporting analytics workflows

PetMap

San Diego, CA

Software Engineer Intern (Front-end Developer)

Oct 2024 - Mar 2025

- Developed and maintained scalable, responsive PetMap web using HTML, Tailwind CSS, JS, and React.js, leading to a 40% increase in user engagement and ensuring cross-device compatibility
- Implemented an interactive Google Maps view using Maps JavaScript API for dynamic visualization of pet locations, handling map initialization and custom marker creation
- Utilized GitHub for version control, delivering over 50+ pull requests, resolving merge conflicts, and coordinating with a team of 4+ developers to deliver consistent, high-quality code updates

Academia Sinica (Institute of Statistical Science)

Taipei, Taiwan

Research Intern

Jul 2024 - Aug 2024

- Developed and conceptualized a research project under the Cancer Moonshot initiative, focusing on innovative strategies to target and analyze cancer-related biomarkers
- Performed correlation analysis (Pearson/Spearman) and data standardization (z-score normalization) and applied multivariate methods (PCA/CCA) to examine relationships between patients' gene expression profiles and phenotypic traits, identifying key cancer-related genetic biomarkers
- Designed and generated detailed visualizations using R, including heatmaps for gene expression patterns, MA plots to assess differential expression, and volcano plots to highlight statistically significant genes

OwlTing Group

Taipei, Taiwan

Software Engineer Intern (Full-stack Developer)

Jul 2023 - Aug 2023

- Developed a dynamic blog website using PHP, incorporating features such as auto-complete search, user and post creation, tagging, commenting, and categorization to enhance user experience and functionality
- Constructed a responsive, visually engaging website using HTML and CSS, ensuring consistent layout across devices
- Enhanced website performance and backend workflows by utilizing Laravel's built-in features, streamlining database interactions, automating common tasks, and accelerating development time by 15%

PROJECT

Loss Table

Nov 2025 - Jan 2026

- Built a platform to calculate total system loss across configurable hardware components, integrating multiple relational tables (loss details, parameters, categories, users) with filtered, sortable views using ASP.NET Core and MSSQL
- Implemented PDF report generation with charts, role-based access control, and Excel-based dropdown maintenance, enabling users to securely export results and manage data